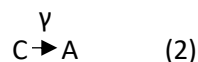
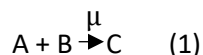


Project 2: Simulating the zombie apocalypse (a.k.a., the spread of ideas)

Please download the research paper (Zombie_Sethna_PRE15.pdf) from Canvas. The theory and model are fairly well described in the paper. For this project, I expect that you will be able to reproduce Figs. 1-4 from the paper. I would also like you to try the 2D spatial simulation and generate something like Fig. 6 (but for a smaller lattice). Can you make a plot that they didn't, which is the average number of zombies on the lattice as a function of α ? It will be like what you observed for $\langle m \rangle$ for the Ising model. The last section where they simulate the spread of an infestation across the US is unfeasible.

To carry out the stochastic simulations of the well-mixed (i.e., no spatial dimension) SZR model, you will need to use the Gillespie algorithm. The paper doesn't provide any details, so here is a brief summary of the algorithm. The Gillespie algorithm is used to model stochastic reaction systems (described by a *master equation*). For a system of particles of types A, B, and C, with associated number of particles N_A , N_B , and N_C , and that react with the following two reactions:



Reaction (1) occurs with rate $r_1 = \mu N_A(t) N_B(t)$, and reaction (2) occurs with rate $r_2 = \gamma N_C(t)$. The total rate of all the reactions is $r_{\text{tot}} = r_1 + r_2$. Since all these are 1st-order processes (depending at most linearly on any of the dynamical variable N_A , N_B , and N_C), the time of the next reaction follows an exponential decay (recall Activity Guide 1). Which reaction occurs then depends on their relative contribution to the total reaction rate. To simulate the reacting system stochastically, Gillespie proposed that we randomly sample a reaction time from an exponential distribution with a rate parameter of r_{tot} and then randomly pick one of the reactions according to their contribution, updating the particle numbers based on the selected reaction. To summarize, the Gillespie algorithm is:

1. Initialize system: set all particle types to their initial numbers, and set time $t = 0$.
2. Calculate the rates r_i of each different reaction. Calculate the total rate $r_{\text{tot}} = \sum_i r_i$ of any reaction
3. Pick a random time dt from the exponential distribution with a rate factor of r_{tot} . Update $t = t + dt$.
4. Randomly pick one of the chemical reactions. Generate a random number on $[0,1)$ and select the chemical reaction using their relative probabilities as $p_i = r_i / r_{\text{tot}}$. (i.e., if one reaction gave a 60% contribution to r_{tot} , then $p_i = 0.6$ and you would have a 60% chance of picking it).
5. Update the particle numbers using the rules for the reaction you randomly selected in 4.
6. Repeat 3-5 until a set amount of time has elapsed.